

Cognitive Thermodynamics, The Ariṣaḍvarga, and the Landauer Veto

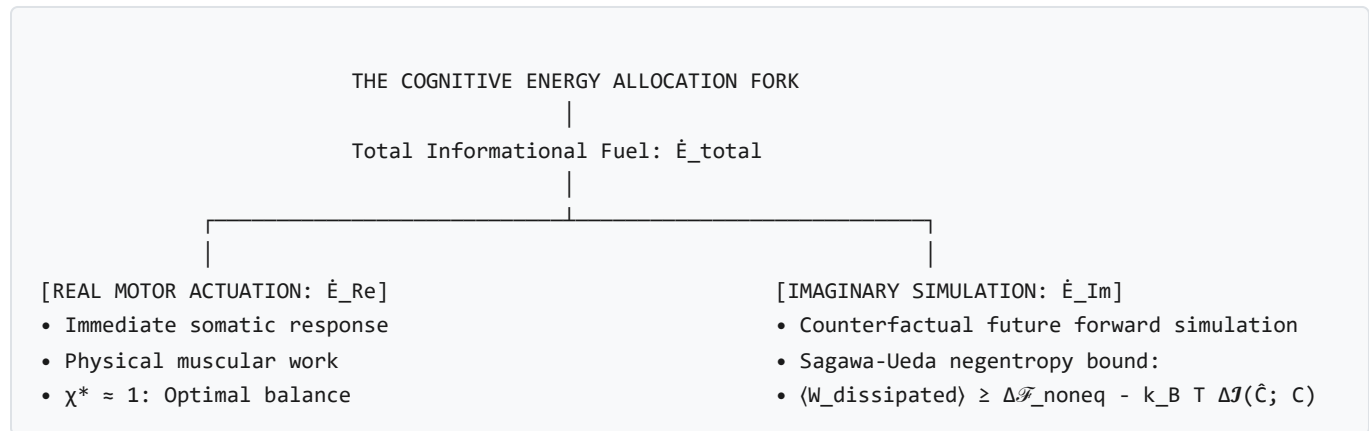
> **Cognitive & Psychological Focus:** This treatise formalizes **Tier III Cognitive Architectures**, **Bayesian Active Inference Engines**, the **Thermodynamics of the Six Internal Afflictions (Ariṣaḍvarga)**, and the **Landauer Information-Erasure Cost of Voluntary Veto (Viveka)**.

1. Tier III: Cognitive Agents as Complex Predictive Engines

In [draft.md` §2.3.4 & §4](../draft.md), a cognitive agent ($E \in T_{III}$) operates across complexified phase space $\Omega_C = \Omega_{\mathbb{R}} \oplus i\Omega_{\mathbb{I}}$:

* **Real Substrate ($\Omega_{\mathbb{R}}$):** Immediate somatic, motor, and neurochemical configurations.

* **Imaginary Substrate ($\Omega_{\mathbb{I}}$):** Internal generative Bayesian models simulating counterfactual future trajectories ($\hat{C}_{\text{future}}$).



The Sagawa-Ueda Information-Thermodynamic Bound:

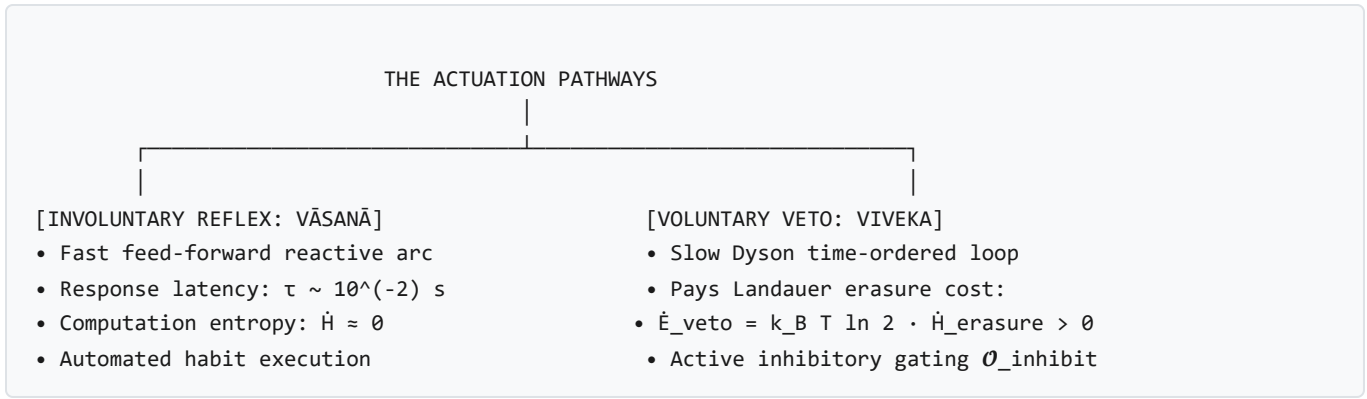
The dissipation required to survive an environmental shock is reduced by the mutual information $\Delta \mathcal{I}(\hat{C}; C_{\text{future}})$ acquired via internal simulation:

$$\langle W_{\text{dissipated}} \rangle \geq \Delta \mathcal{F}_{\text{noneq}} - k_B T \cdot \Delta \mathcal{I}(\hat{C}_{\text{predicted}}; C_{\text{future}})$$

Pre-stiffening cognitive priors ($\mathbf{R}_{\text{active}}(t - \Delta t)$) reduces physical damage to near zero upon predictable impacts.

2. Involuntary Latency (Vāsanā) vs. Voluntary Landauer Veto Gating (Viveka)

Living cognitive agents possess two distinct modes of interfacial actuation:



1. Involuntary Reflex (*Vāsanā* / *Saṃskāra*):

An automated feed-forward reflex arc where incoming challenge traction $\mathbf{C}$ directly triggers motor expression $\mathcal{X}$ through pre-configured neural pathways ($\tau_{\text{latency}} \sim 10^{-2}$ s). Because no deliberative computation or state re-evaluation occurs, the Landauer informational entropy cost is zero ($\dot{\mathcal{H}} \approx 0$).

2. Voluntary Discriminative Veto (*Viveka*):

The conscious intervention of the higher-order Dyson loop ($\mathcal{T} \exp(\int \hat{\mathcal{L}} d\tau)$), which inspects the automated trajectory and applies an active inhibitory gate ($\mathcal{O}_{\text{inhibit}}$) to prevent expression.

* **The Landauer Metabolic Tax:** Erasing the automated motor command requires physical bit erasure, demanding continuous free-energy dissipation:

$$\dot{\mathcal{E}}_{\text{veto}} = k_B T \ln 2 \cdot \dot{\mathcal{H}}_{\text{erasure}} > 0$$

Exercising conscious restraint (Viveka / Dama)* is physically exhausting because it consumes real biochemical glucose ($k_B T \ln 2$) to clear neural register buffers.

3. Top-Down Craving (*Kāma*) vs. Bottom-Up Hunger (*Kṣudhā*)

Hunger (*Kṣudhā**):

A bottom-up First-Law exergy deficit occurring entirely in real configuration space $\Omega_{\mathbb{R}}$:

$$\dot{\mathcal{E}}_{\text{fuel}} < \dot{\mathcal{E}}_{\text{crit}} \equiv T_{\text{ambient}} \int_E \sigma_{\text{total}} dV \implies \frac{d\mathcal{G}}{dt} < 0$$

Kṣudhā is a biochemical indicator of metabolic depletion (falling blood glucose, low ATP/AMP ratio) requiring zero memory simulation.

Desire / Craving (*Kāma* / *Trṣṇā**):

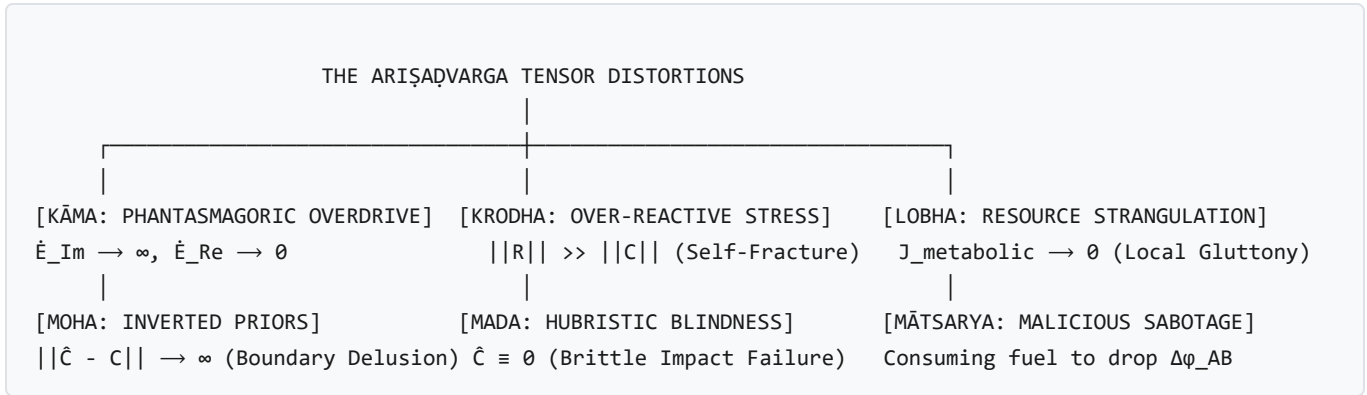
A top-down simulation executed in imaginary phase space $\Omega_{\mathcal{I}\mathbb{m}}$:

$$D_{\mathcal{I}\mathbb{m}} = \hat{\pi}(\mathcal{F}_{\text{ledger}}) \xrightarrow{\text{simulation}} \Delta \hat{\mathcal{G}}_{\text{predicted}} > 0$$

Kāma occurs when the cognitive model recalls past reward states from $\mathcal{F}_{\text{ledger}}$ and projects an artificial free-energy surplus. This top-down prediction transduces downward into physical physiology via the Semantic Transduction Tensor ($\mathbf{C}_{\text{craving}} = \mathbf{K}_{\text{trans}} \cdot \nabla_{\theta} \mathcal{D}_{\text{KL}}$), generating visceral dopamine cascades and restlessness **even when fully satiated**.

4. Thermodynamics of the *Ariṣaḍvarga* (The Six Internal Afflictions)

In classical Sanskrit psychology, the *Ariṣaḍvarga* represents the six inner enemies that destroy discernment. In our framework, these are derived as **exact mathematical distortions of the structural margin tensor** $\phi(x, t)$:



1. Kāma (काम — Unbounded Imaginary Projection Overdrive):

Informational fuel allocation diverges ($\chi \rightarrow \infty \implies \dot{E}_{\text{Im}} \rightarrow \dot{E}_{\text{total}}, \dot{E}_{\text{Re}} \rightarrow 0$). The entity expends all free energy simulating ungrounded imaginary prospects while the physical substrate starves into anergic boundary collapse.

2. Krodha (क्रोध — Explosive Over-Reactive Stress Discharge):

Internal counter-stress vastly exceeds external challenge ($||\mathbf{R}_{\text{active}}|| \gg ||\mathbf{C}||$). The localized stress concentration generates internal Rankine-Hugoniot shock waves that fracture the entity's own boundary from within.

3. Lobha (लोभ — Parasitic Resource Hoarding):

A constituent node hoards fuel measure ($\mu(E^j) \uparrow$) while choking metabolic syncytial flux ($J_{\text{metabolic}}^{j \rightarrow k} \rightarrow 0$). This starves neighboring nodes, triggering a localized ischemia cascade that collapses the macro-envelope.

4. Moha (मोह — Perceptual Delusion / Model Inversion):

The generative model divergence explodes ($||\hat{C} - C_{\text{true}}|| \rightarrow \infty$). The entity mistakes destructive threats for fuel sources, actively opening its boundary pores ($\mathbf{v}_n \cdot \hat{n} < 0$) to toxic foreign challenge fields.

5. Mada (मद — Hubristic Invulnerability / Zero-Challenge Prior):

The entity sets its predicted external challenge to zero ($\hat{C} \equiv 0$), disarming its adaptive viscoelastic dampers ($\nu \rightarrow 0$). Upon the arrival of real environmental shocks, it undergoes catastrophic brittle fracture without damping.

6. Mātsarya (मात्सर्य — Malicious Zero-Sum Trophic Sabotage):

An entity expends its own metabolic exergy solely to diminish a neighbor's margin ($\Delta\phi_{AB} < 0$), lowering total syncytial free energy and precipitating mutual systemic lysis.